# 1. PROJECT DESCRIPTION

# [Project Name]

**SmartQR CMS – Restaurant, Dine-in & Online Ordering System**

# [Project Description]

SmartQR CMS is a web-based platform that allows restaurants to manage their menus, tables, and orders through a QR code ordering system. Customers can scan a QR code placed on each table to access the restaurant's digital menu, place orders directly from their mobile devices, and complete payment online.

The system is designed as a **Software-as-a-Service (SaaS)** solution, allowing multiple restaurants (tenants) to use the platform independently. Each restaurant can manage its own staff, menu items, tables, and orders through an administrative dashboard. Kitchen staff and waiters can view and manage incoming orders in real time, improving efficiency and reducing manual errors in the ordering process.

By integrating **QR technology, online payment, and real-time order management**, the system enhances the dining experience for customers while helping restaurants streamline operations and improve service quality.

# [Project Purpose]

The purpose of this project is to develop a digital ordering system that simplifies the ordering and payment process in restaurants. Traditional ordering methods often rely on manual interaction between customers and waiters, which can lead to delays, order mistakes, and inefficient service.

This system aims to replace or support the traditional workflow by allowing customers to order directly through their smartphones by scanning a QR code. At the same time, restaurant staff can manage orders, menus, and tables more efficiently using a centralized system.

Additionally, the project aims to provide a scalable SaaS platform where multiple restaurants can use the system without needing to develop their own custom software solutions.

# [Project Goals]

The main goals of the project include:

1. To develop a QR-based ordering system that allows customers to scan a table QR code and access the restaurant menu.
2. To enable customers to place orders and complete online payments directly from their mobile devices.
3. To provide restaurant staff (waiters and chefs) with tools to manage and process orders efficiently.
4. To allow restaurant administrators to manage menus, tables, and staff through an administrative dashboard.
5. To support a multi-tenant SaaS architecture so multiple restaurants can use the system independently.
6. To improve restaurant operational efficiency and enhance the customer dining experience.

# [Project Objectives]

## 1. Admin & Tenant Admin

*Menu Management*

1. Create Category
2. Update Category
3. Delete Category
4. Create Menu Item
5. Update Menu Item
6. Delete Menu Item
7. Upload Menu Image
8. Enable / Disable Menu Item

*Table Management*

9. Create Table
10. Update Table
11. Delete Table
12. Generate QR Code for Table

*Staff Management*

13. Create Staff
14. Update Staff
15. Delete Staff
16. Assign Staff Role (Waiter / Chef)

*Order & Report Management*

17. View Orders
18. View Order History
19. View Sales Report
20. View Revenue Statistics

## 2. Waiter

- View Orders
- View Orders by Table
- Update Order Status
- Serve Order
- Create Order for Customer (manual)

## 3. Chef

- View Pending Orders
- View Order Details
- Update Cooking Status
- Mark Order as Ready

## 4. Customer

1. Scan Table QR
2. View Menu
3. View Menu by Category
4. View Menu Item Detail
5. Add Item to Cart

6. Update Cart (change quantity / remove item)
7. Checkout
8. Make Payment
9. View Payment Result
10. Track Order Status

# [Project Scope]

## Customer Functions

- Scan QR code from the table
- View digital menu
- Add items to cart
- Place orders
- Make online payments
- Track order status

## Restaurant Staff Functions

- Waiters can view and manage orders
- Chefs can view incoming orders and update cooking status

## Restaurant Admin Functions

- Manage menu categories and menu items
- Manage restaurant tables
- Generate QR codes for tables
- Manage staff accounts and roles
- View order history and sales reports

## System Features

- QR code generation for tables
- Multi-tenant restaurant management
- Real-time order management
- Online payment integration

The project does **not include** hardware systems such as POS machines or kitchen display hardware, but it can integrate with them in future development.

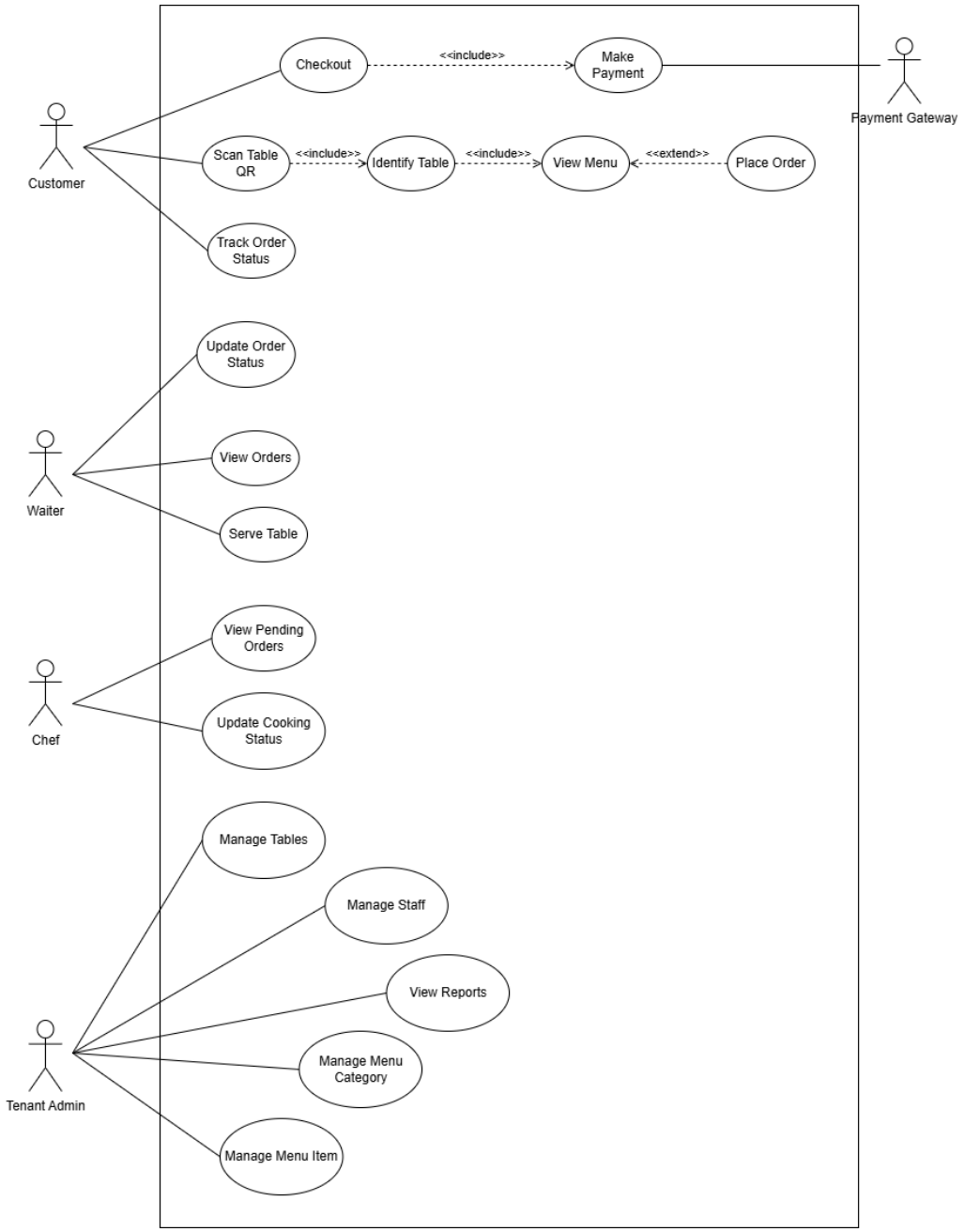
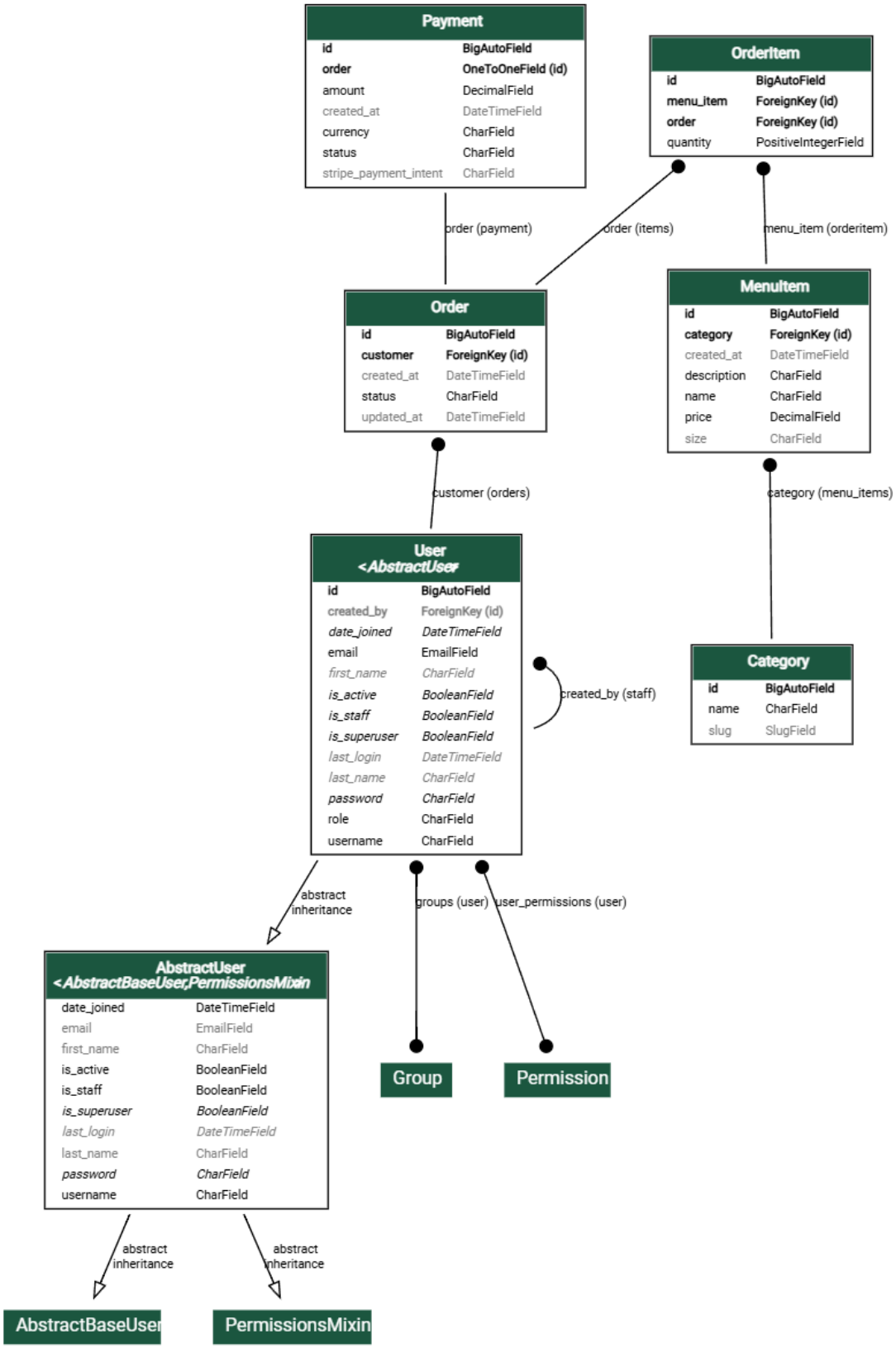
# [Project Deliverables]

The main deliverables of this project include:

1. **System Documentation**
    - Software Requirements Specification (SRS)U'
    - System Design Documents
    - Use Case Diagrams
    - Database Design (ERD)
2. **Backend System**
    - RESTful API developed using **Django REST Framework**
    - Database models for users, menus, tables, orders, and payments
3. **Frontend Interface**
    - Web interface for customers to view menu and place orders
    - Admin dashboard for restaurant management
4. **QR Code Ordering System**
    - QR codes generated for each table
    - QR scanning to access the digital menu
5. **Payment Integration**
    - Online payment processing for customer orders
6. **Testing and Deployment**
    - System testing
    - Deployment-ready web application

# 2. Order Process

# 3. USE-CASE DIAGRAM



# 4. ERD (version 1, still in progress)